

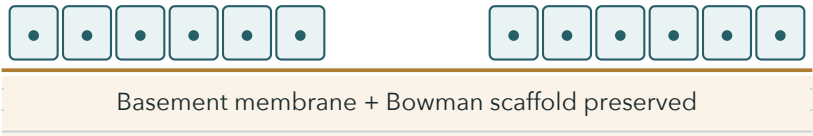
Corneal barrier and healing

Depth of injury predicts the repair route and the risk of an optical scar.

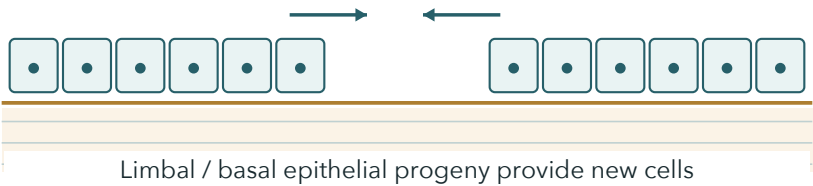
A superficial barrier can renew

Closing a surface defect

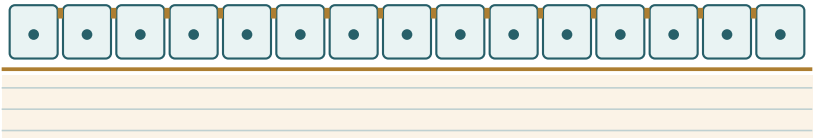
1 SURFACE DEFECT



2 CELLS MIGRATE, THEN REPLENISH



3 JOINED SURFACE BARRIER



Junctions close the extracellular entry route

Superficial abrasion with preserved underlying scaffold. The sequence is simplified; migration and proliferation can overlap.

Barrier, attachment, replacement

Component	Role
Surface-cell junctions	Restrict passage between adjacent epithelial cells.
Basement membrane	Supports attachment and regulates epithelial-stromal signalling.
Limbal progenitors	Supply renewing corneal epithelial cells.

Use the dye logically

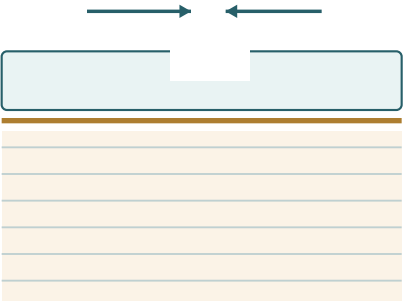
Fluorescein highlights an epithelial defect. Loss of staining after closure indicates surface repair; it does not prove that a deeper stromal scar has cleared.

Surface closure versus optical recovery

The depth switch

SUPERFICIAL

INTO STROMA

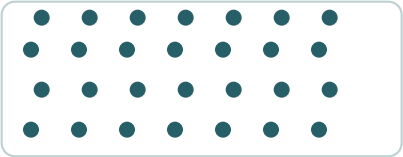


Cells close the defect;
stromal order preserved

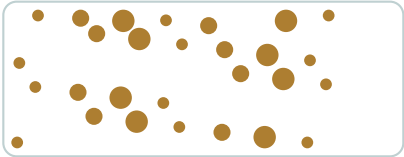


Bowman disrupted;
stromal matrix remodels

STROMAL COLLAGEN



Uniform size / spacing



Disordered scar: scatter

A wound entering stroma may heal with disordered matrix. The illustrated collagen fields show optical order, not exact fibril dimensions.

Depth changes the outcome

Injury	Expected reasoning
Epithelium only	Regeneration can restore a clear surface over an intact scaffold.
Bowman + anterior stroma	Bowman does not normally regenerate; stromal fibrosis can scatter light.
Persistent basement-membrane damage	Profibrotic signalling can sustain myofibroblasts and haze.

Why normal cornea transmits light

Uniform collagen fibril diameter, controlled spacing and avascularity limit scatter. Endothelial hydration control is another requirement, not a substitute for stromal order.